Calculus II - AY 2024-25

Chapter 4. Second-order Ordinary Differential Equations

Outline

1. Some Simple Forms of Second-Order Equations
2. Second-order Linear Differential Equations
3. Homogeneous Lin. Diff. Eq. with Constant-Coefficients
4. The Method of Undetermined Coefficients
5. The Method of Variation-of-Parameters
6. Series Solutions to Linear Differential Equations

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Some Simple Forms of Second-Order Equations

Discussion

The **methods** we have discussed in **Chapter 3** are only for solving special types of **first-order** differential equations, and do not apply directly to higher-order equations.

The solution to such equations usually requires the derivation of new techniques.

One possible approach is to **replace** a higher-order differential equations by an equivalent **system** of first-order equations.

For example, any second-order differential equation that can be written in the form

$$\frac{d^2y}{dx^2} = F\left(x, y, \frac{dy}{dx}\right), \tag{1}$$

where F is a known function, can be replaced by an equivalent system of first-order differential equations as follows.

Let $z = \frac{dy}{dx}$. Then $\frac{d^2y}{dx^2} = \frac{dz}{dx}$, and so solving Equation (1) is equivalent to solving the following two first-order differential equations

$$\begin{cases} \dfrac{dy}{dx} = z & (2) \\ \dfrac{dz}{dx} = F(x, y, z). & (3) \end{cases}$$

Equation (3) $\dfrac{dz}{dx} = F(x, y, z)$, in general, cannot be solved directly, since it involves three variables x, y, and z.

However, for **certain forms** of the function F, this equation will involve only **two** variables, and then it can sometimes be solved for z using techniques presented in ODE (Chapter 3).

Having obtained z, one can then substitute into Equation (2) $\dfrac{dy}{dx} = z$ to obtain a first-order differential equation for y.

Now we discuss **two forms** of F for which this is certainly the case.

Problem (second-order differential equations with the **dependent variable y missing**)

If y does **not** occur explicitly in the function F, then Equation (1) has the form

$$\frac{d^2y}{dx^2} = F\left(x, \frac{dy}{dx}\right). \tag{4}$$

Since $z = \dfrac{dy}{dx}$ and $\dfrac{dz}{dx} = \dfrac{d^2y}{dx^2}$, this equation can be replaced by two first-order differential equations

$$\begin{cases} \dfrac{dy}{dx} = z & (5) \\ \dfrac{dz}{dx} = F(x, z). & (6) \end{cases}$$

Thus to solve Equation (4), we first solve Equation (6) for z in terms of x and then solve Equation (5) for y as a function of x.

Examples

Find the general solution to

$$\frac{d^2y}{dx^2} = \frac{1}{x}\left(\frac{dy}{dx} + x^2\cos x\right), \quad x > 0.$$

Solution

In the given equation the dependent variable y **is missing**. We have the following equivalent system of first-order differential equations

$$\begin{cases} \dfrac{dy}{dx} = z \\ \dfrac{dz}{dx} = \dfrac{1}{x}(z + x^2\cos x). \end{cases}$$

The second equation is a first-order linear differential equation

with standard form $\dfrac{dz}{dx} - \dfrac{1}{x}z = x\cos x$, which has an integrating factor $I(x) = e^{-\int \frac{1}{x}\,dx} = \dfrac{1}{x}$.

Hence

$$\frac{d}{dx}\left(\frac{1}{x}z\right) = \cos x \Longrightarrow \frac{1}{x}z = \sin x + C \Longrightarrow z = x\sin x + Cx.$$

From

$$\frac{dy}{dx} = x\sin x + Cx$$

it follows that

$$y = -x\cos x + \sin x + C_1 x^2 + C_2,$$

where C_1, C_2 are arbitrary constants.

Problem (second-order differential equations with the **independent variable x missing**)

If x does **not** occur explicitly in the function F in Equation (1), then we must solve a differential equation of the form

$$\frac{d^2y}{dx^2} = F\left(y, \frac{dy}{dx}\right). \tag{7}$$

As in the previous case, let $z = \dfrac{dy}{dx}$, by the chain rule

$$\frac{d^2y}{dx^2} = \frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx} = z\frac{dz}{dy}.$$

Substituting for $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ into Equation (7) reduces the second-order differential equation to the equivalent first-order system

$$\begin{cases} \frac{dy}{dx} = z, & (8) \\ z\frac{dz}{dy} = F(y, z). & (9) \end{cases}$$

Thus to solve Equation (7), first solve Equation (9) for z in terms of y and then solve Equation (8) for y as a function of x.

Examples

Find the general solution to

$$\frac{d^2y}{dx^2} = -\frac{2}{1-y}\left(\frac{dy}{dx}\right)^2.$$

Solution

In the given differential equation the independent variable x **is missing**. We have the following equivalent system

$$\begin{cases} \dfrac{dy}{dx} = z \\ z\dfrac{dz}{dy} = -\dfrac{2}{1-y}z^2. \end{cases}$$

For the second equation, if separate the variables, we have to assume that $z \neq 0$. So we consider two cases:

- If $z = 0$: then the first equation gives $y = \text{constant}\ (\neq 1)$, which is a solution of the given differential equation.
- For $z \neq 0$ (and hence y is non-constant), separating the variables we obtain

$$\frac{1}{z}\, dz = -\frac{2}{1-y}\, dy \Longrightarrow \log|z| = 2\log|1-y| + C.$$

From the last equation, it follows that

$$z = C_1(1-y)^2,$$

where we have set $C_1 = \pm e^C$, and so $C_1 \neq 0$.

Hence

$$\frac{dy}{dx} = C_1(1-y)^2.$$

Separating and integrating (remember $y \neq 1$, by the given equation), we obtain

$$\frac{1}{1-y} = C_1 x + C_2,$$

or

$$y = \frac{C_1 x + (C_2 - 1)}{C_1 x + C_2},$$

where $C_1 \neq 0$ and C_2 are arbitrary constants.

Summarizing the obtained results, we get two solutions

$$y = C; \quad y = \frac{C_1 x + (C_2 - 1)}{C_1 x + C_2},$$

here $C \neq 1$, $C_1 \neq 0$ and C_2 are constants.

Note that the first solution is contained in the second one (when $C_1 = 0$), and so the general solution to the given differential equation can be written as follows

$$y = \frac{C_1 x + (C_2 - 1)}{C_1 x + C_2},$$

where C_1 and C_2 are arbitrary constants.

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Second-order Linear Differential Equations

The second-order linear differential equations have important applications, especially in mechanical and electrical engineering.

The **solution techniques** for second-order linear differential equations are quite **simple** (in comparison with those of higher order).

A second-order **linear** differential equation is an equation of the form

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = F(x).$$

Here a_0, a_1, a_2 and F are functions defined on an interval I.

Assume that $a_0(x) \neq 0$ on I. We divide both side of the equation by $a_0(x)$ to obtain the **standard form**

$$y'' + a(x)y' + b(x)y = F(x). \tag{1}$$

For studying this equations, it is needed to consider separately two equations:

1) The **homogeneous** equation associated to (1)

$$y'' + a(x)y' + b(x)y = 0.$$

2) The **non-homogeneous** equation (1) itself

$$y'' + a(x)y' + b(x)y = F(x).$$

Now we introduce the following notion which is important for further study.

Definition

Two functions defined on the interval I are said to be **linearly dependent** on I, if one function is a scalar multiple of another function.

Otherwise, these functions are said to be **linearly independent** on I.

This is precisely what we have studied for the vector space $\mathcal{F}(I, \mathbb{R})$ in Linear Algebra part.

Example

Two functions $\sin x$ and $\cos x$ are *linearly independent* for all x.

Solution

It's easy (!)

Wronskian

In practice, for homogeneous linear second-order differential equations, the following result is useful for verifying whether two solutions are linearly dependent/independent.

Theorem

Let y_1 and y_2 be two (differentiable) solutions to a homogeneous linear second-order differential equation $y'' + a(x)y' + b(x)y = 0$ on the interval I. Consider the (variable) determinant

$$W[y_1, y_2] = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} \quad (x \in I),$$

called the **Wronskian** *of y_1 and y_2.*

Theorem (cont'd)

1) *If* $W[y_1, y_2](x) \neq 0$ *at* <u>some</u> *point* $x_0 \in I$*, then solutions* y_1 *and* y_2 *are linearly independent on* I*.*
2) *If* $W[y_1, y_2](x) \equiv 0$ *on* I*, then solutions* y_1 *and* y_2 *are linearly dependent on* I*.*

Note that in the theorem above, (2) is applied for two solutions of the homogeneous linear second-order differential equation

$$y'' + a(x)y' + b(x)y = 0$$

It is not true for a general case of functions, that is if the Wronskian of two functions is identically zero on some interval, there is no information on whether these two functions are linearly dependent or linearly independent.

Exercise

Provide counterexamples to show that $W[f_1, f_2](x) \equiv 0$ on some interval I, but the set $\{f_1, f_2\}$ may or may not be linearly dependent.

Example

Verify that $y_1(x) = e^x$ and $y_2(x) = xe^x$ are solutions to the differential equation $y'' - 2y' + y = 0$. Determine whether they are linearly dependent or linearly independent on $I = (-\infty, \infty)$.

Solution

First, we can easily check (!) that $y_1(x) = e^x$ and $y_2(x) = xe^x$ are solution to the given equation.

Furthermore, since

$$W[y_1, y_2](x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} \neq 0,$$

for all $x \in (-\infty, \infty)$, the solutions are linearly independent on I.

Example

Verify that $y_1(x) = \cos 2x$ and $y_2(x) = 3(1 - 2\sin^2 x)$ are solutions to the differential equation $y'' + 4y = 0$.

Determine whether they are linearly dependent or linearly independent on $I = (-\infty, \infty)$.

Solution

First, we can easily check (!) that $y_1(x) = \cos 2x$ and $y_2(x) = 3(1 - 2\sin^2 x)$ are solution to the given equation.

Furthermore, since

$$W[y_1, y_2](x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & 3(1 - 2\sin^2 x) \\ -2\sin 2x & -12\sin x \cos x \end{vmatrix}$$
$$= -6\cos 2x \sin 2x + 6\sin 2x \cos 2x \equiv 0,$$

for all $x \in (-\infty, \infty)$,the solutions are linearly dependent on I.

Indeed, for the example above, since $\cos 2x = 1 - 2\sin^2 x$, the dependency relation is $3y_1 - y_2 = 0$. We can verify that a second linearly independent solution to the given differential equation is $y_3(x) = \sin 2x$.

Linear differential equations have a rich solution structure.

Especially, for homogeneous equations a great advantage of this structure is given in the following theorem, which sometimes is called the **linearity principle**.

Theorem

For a homogeneous linear differential equation

$$y'' + a(x)y' + b(x)y = 0,$$

if $y_1(x)$ and $y_2(x)$ are two solutions on the interval I, then any their linear combination $y(x) = C_1 y_1(x) + C_2 y_2(x)$, where C_1, C_2 are constants, is again a solution on I.

In particular, for such an equation, sums and scalar multiples of solutions are again solutions.

Proof

Easy (!)

Remark.

The linearity principle holds for **only** homogeneous linear differential equations, but **does not** hold for non-homogeneous (linear or non-linear) differential equations.

Example

For the non-homogeneous linear differential equation

$$y'' + y = 1,$$

the functions $y_1(x) = 1 + \cos x$ and $y_2(x) = 1 + \sin x$ are solutions, but their sum is not a solution. Neither is $C(1 + \cos x)$ or $C(1 + \sin x)$ with any $C \neq 1$.

Example

For the non-linear differential equation

$$y''y - xy' = 0,$$

the functions $y_1(x) = 1$ and $y_2(x) = x^2$ are solutions, but their sum is not a solution. Neither is Cx^2 with any $C \neq 0, \neq 1$.

Now we consider the (standard) second-order non-homogeneous linear differential equation

$$y'' + a(x)y' + b(x)y = F(x),$$

where $a(x), b(x), F(x)$ are continuous, and $F(x)$ is not identically zero on the interval I.

Definition

The equation

$$y'' + a(x)y' + b(x)y = 0$$

is called the **associated homogeneous equation** to the non-homogeneous equation.

The main results for non-homogeneous linear differential equations and their associated homogeneous equations are given in the following theorem.

Theorem

1) The **general solution** of the equation $y'' + a(x)y' + b(x)y = 0$ is of the form

$$y(x) = C_1 y_1(x) + C_2 y_2(x),$$

where $y_1(x)$ and $y_2(x)$ are two linearly independent solutions to the given equation, and C_1 and C_2 are arbitrary constants.

Theorem (cont'd)

2) • $y_1(x)$ and $y_2(x)$ are two linearly independent solutions to the homogeneous equation $y'' + a(x)y' + b(x)y = 0$ on an interval I

• $y_p(x)$ is any **particular** solution to the non-homogeneous equation $y'' + a(x)y' + b(x)y = F(x)$ on I.

Then every solution to $y'' + a(x)y' + b(x)y = F(x)$ on I is of the form

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x),$$

for appropriate constants C_1, C_2.

Thus the **general solution** to

$$y'' + a(x)y' + b(x)y = F(x)$$

is of the form

$$y(x) = y_c(x) + y_p(x),$$

where

$$y_c(x) = C_1 y_1(x) + C_2 y_2(x)$$

is the general solution to $y'' + a(x)y' + b(x)y = 0$, and y_p is a particular solution to $y'' + a(x)y' + b(x)y = F(x)$.

We refer y_c as the **complementary function** for $y'' + a(x)y' + b(x)y = F(x)$.

We need one final result regarding solutions to non-homogeneous differential equations, whose proof is rather obvious.

Theorem

If u_p and u_q are particular solutions of

$$y'' + a(x)y' + b(x)y = f(x) \quad \text{and} \quad y'' + a(x)y' + b(x)y = g(x),$$

respectively, then $u_p + u_q$ is a solution to

$$y'' + a(x)y' + b(x)y = f(x) + g(x).$$

In the sequel, we study only those linear differential equations, whose coefficients are **constants**.

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Homogeneous Lin. Diff. Eq. with Constant-Coefficients

Consider a homogeneous linear differential equation of order 2 with constant coefficients

$$y'' + ay' + by = 0,$$

where a, b are **constants**.

The **solution technique** for solving this equation: Define the quadratic equation

$$P(t) = t^2 + at + b = 0,$$

which is called the **characteristic equation** of the given homogeneous differential equation.

The roots of this quadratic equation are

$$t_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}.$$

There are three possibilities for the roots, depending on the sign of the discriminant $\Delta = a^2 - 4b$:

1) If $\Delta > 0$, then $P(t)$ has two distinct real roots.
2) If $\Delta = 0$, then $P(t)$ has a real double root.
3) If $\Delta < 0$, then $P(t)$ has two complex conjugate roots.

* **Case 1: t_1, t_2 are distinct real roots**

In this case, two linearly independent solutions are

$$y_1(x) = e^{t_1 x}, \; y_2(x) = e^{t_2 x},$$

and the corresponding general solution is

$$y(x) = C_1 e^{t_1 x} + C_2 e^{t_2 x},$$

where C_1, C_2 are arbitrary constants.

Example

Determine the general solution to

$$y'' - y' - 2y = 0.$$

Solution.

We have $P(t) = t^2 - t - 2 = (t-2)(t+1) = 0 \quad \Longrightarrow t_1 = 2, t_2 = -1.$
Then two linearly independent solutions to the given differential equation are $y_1(x) = e^{2x}$ and $y_2(x) = e^{-x}$. Therefore, the general solution to the differential equation is

$$y(x) = C_1 e^{-x} + C_2 e^{2x},$$

where C_1, C_2 are arbitrary constants.

* **Case 2: $t_1 = t_2$ are a real double root t_0**

In this case, two linearly independent solutions are

$$y_1(x) = e^{t_0 x}, \; y_2(x) = xe^{t_0 x},$$

and the corresponding general solution is

$$y(x) = C_1 e^{t_0 x} + xC_2 e^{t_0 x},$$

where C_1, C_2 are arbitrary constants.
(Here $t_0 = -\frac{a}{2}$.)

Example

Determine the general solution to

$$y'' + 4y' + 4y = 0.$$

Solution.

We have $P(t) = t^2 + 4t + 4 = (t+2)^2 = 0 \quad \Longrightarrow t_1 = t_2 = -2.$
Then two linearly independent solutions to the given differential equation are $y_1(x) = e^{-2x}$ and $y_2(x) = xe^{-2x}$. Therefore, the general solution to the differential equation is

$$y(x) = C_1 e^{-2x} + C_2 x e^{-2x},$$

where C_1, C_2 are arbitrary constants.

* **Case 3: $t_{1,2} = \alpha \pm i\omega$ are two complex conjugate roots**

In this case, two linearly independent solutions are

$$y_1(x) = e^{\alpha x}\cos\omega x,\ y_2(x) = e^{\alpha x}\sin\omega x,$$

and the corresponding general solution is

$$y(x) = C_1 e^{\alpha x}\cos\omega x + C_2 e^{\alpha x}\sin\omega x,$$

where C_1, C_2 are arbitrary constants.
(Here $\alpha = -\dfrac{a}{2},\ \omega = \dfrac{\sqrt{-\Delta}}{2}$.)

Example

Determine the general solution to

$$y'' + 6y' + 25y = 0.$$

Solution.

We have $P(t) = t^2 + 6t + 25 = 0 \quad \Longrightarrow t_1 = -3 + 4i, t_2 = -3 - 4i.$
Then two linearly independent solutions to the given differential equation are $y_1(x) = e^{-3x}\cos 4x$ and $y_2(x) = e^{-3x}\sin 4x$. Therefore, the general solution to the differential equation is

$$y(x) = C_1 e^{-3x}\cos 4x + C_2 e^{-3x}\sin 4x,$$

where C_1, C_2 are arbitrary constants.

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In this section we study the technique for obtaining a particular solution y_p to the equation

$$y'' + ay' + by = F(x),$$

when $F(x)$ is one of the following forms:

- a power of x
- an exponential function
- a sine or cosine
- sums or products of such functions.

The reason is that these functions have derivatives similar to $F(x)$ itself.

The **technique** can be introduced quite simply as follows: choose a form for y_p similar to $F(x)$, but with unknown coefficients.

These coefficients will be determined by substituting that y_p into the given equation. That is why this is called the **method of undetermined coefficients**.

The choice rules in this method are as follows:

(a) **Basic rule:**

1) If $F(x) = Cx^k$ $(k = 0, 1, \ldots)$, then choose $y_p(x) = A_0 + A_1 x + \cdots + A_k x^k$.
2) If $F(x) = Ce^{\alpha x}$, then choose $y_p(x) = Ae^{\alpha x}$.
3) If $F(x) = C \sin \omega x$ or $F(x) = C \cos \omega x$, then choose $y_p(x) = A_0 \sin \omega x + A_1 \cos \omega x$.

(b) **Sum/product rule:** If $F(x)$ is sums or products of equations 1) – 3), then $y_p(x)$ is sums or products of corresponding functions in those equations.

Example

Solve the initial-valued problem

$$y'' - y' - 2y = 10 \sin x, \quad y(0) = 0, y'(0) = 1.$$

Solution

We have

$$P(t) = t^2 - t - 2 = (t-2)(t+1) = 0 \Longrightarrow t_1 = 2, t_2 = -1.$$

Then the complementary function is

$$y_c(x) = C_1 e^{2x} + C_2 e^{-x}.$$

Furthermore, since $F(x) = 10 \sin x$, we choose

$$y_p(x) = A_0 \sin x + A_1 \cos x.$$

Substituting this function into the given equation, we obtain

$$(-A_0 \sin x - A_1 \cos x) - (A_0 \cos x - A_1 \sin x)$$

$$-2(A_0 \sin x + A_1 \cos x) = 10 \sin x.$$

That is,

$$(-3A_0 + A_1) \sin x - (A_0 + 3A_1) \cos x = 10 \sin x.$$

This equation is satisfied for all x if and only if

$$-3A_0 + A_1 = 10,\ A_0 + 3A_1 = 0,$$

which gives

$$A_0 = -3,\ A_1 = 1.$$

Thus a particular solution is

$$y_p(x) = -3\sin x + \cos x.$$

Consequently the general solution is

$$y(x) = C_1 e^{2x} + C_2 e^{-x} - 3\sin x + \cos x,$$

where C_1, C_2 are arbitrary constants.

Imposing the initial conditions on this solution yields

$$C_1 + C_2 = -1,\ 2C_1 - C_2 = 4 \quad \Longrightarrow \quad C_1 = 1,\ C_2 = -2.$$

Thus the answer is

$$y(x) = e^{2x} - 2e^{-x} - 3\sin x + \cos x.$$

□